

Registration Number: 23BEC10081

Email:rupali.23bec10081@vitbhopal.ac.in

Phone: +91 8629930500

<https://www.linkedin.com/in/rupali-patel-2483a72a7>

Rupali Patel

Technical Skills: Python, Fundamental in AI and ML, C++, Java, MATLAB Programming

Certification:

- Python completion certificate, 27 December 2023
- Certificate of Electro VLSI , 19th - 20th October 2023
- Coursera Certificate of Applied Machine Learning in Python, 23 November 2024

ACADEMIC PROJECTS	
Object Detection In Water Sep 2024- Dec 2024	- Developed and tested a sonar-based object detection system optimized for underwater environments using Arduino and a JSN sensor. - Calibrated the system to adapt to water conditions, including variations in density and sound propagation, ensuring accurate detection. - Potential applications include underwater robotics, autonomous marine vehicles, and disaster recovery operations. - Demonstrated a cost-effective alternative to commercial underwater detection systems, suitable for marine exploration and environmental monitoring.
Sign Language to Speech Conversion Jan 2025-Apr 2025	- Designed and implemented a hardware-based system to convert hand gestures of sign language into audible speech using Arduino Uno. - Utilized flex sensors and an accelerometer to capture finger bends and hand movements, enabling accurate gesture recognition. - Programmed the Arduino to interpret sensor data and map it to corresponding alphabets or words in real time. - Integrated a speech synthesis module to convert recognized gestures into spoken output, facilitating communication for speech and hearing-impaired users. - Built a low-cost, portable prototype with potential applications in assistive communication devices and inclusive technology solution

Education			
Board	Tenure	Educational Institution	CGPA/Percentage
B. Tech (ECE)	Oct 23- Ongoing	Vellore Institute of Technology, Bhopal	9.01/10
Class XII	May 23	Govt. higher secondary school Sodulpur Harda(MP)	91.8%
Class X	July 21	Saraswati Shishu Vidhya mandir, Sodalpur, Harda(MP)	63%

Extra- Curricular And Achievements	
Achievements	- Awarded with Dreamed STAR Scholarship from VIT Bhopal - Awarded with Reliance Foundation Scholarship for academic excellence - Secured position in district merit in class 12th
Extracurricular Activities	- VIT Bhopal's ADVITYA

Additional Information	
Hobbies	- Gaming, Reading books, Sketching